5 **(a)** A battery of e.m.f. E and negligible internal resistance is connected to a NTC thermistor of resistance R_{TH}. In addition, resistor X of resistance R_X is connected in series to the thermistor, as shown in Fig. 5.1.

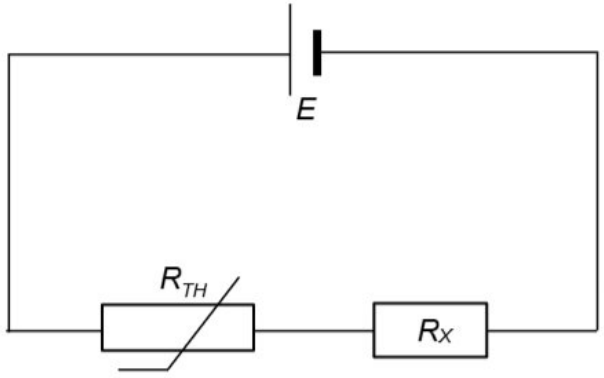


Fig. 5.1

(i) Express the power P_X dissipated in X in terms of E, R_X and R_{TH}.

P_X = ... [2]

(ii) At room temperature, R_{TH} is measured to be double that of R_X.

State and explain how P_X will change if the temperature of the thermistor decreases.

.....

.....

.....

..... [2]

(iii) A secondary circuit with another battery of e.m.f. E_S (of negligible internal resistance) and a galvanometer is connected in parallel to the thermistor, as shown in Fig. 5.2.

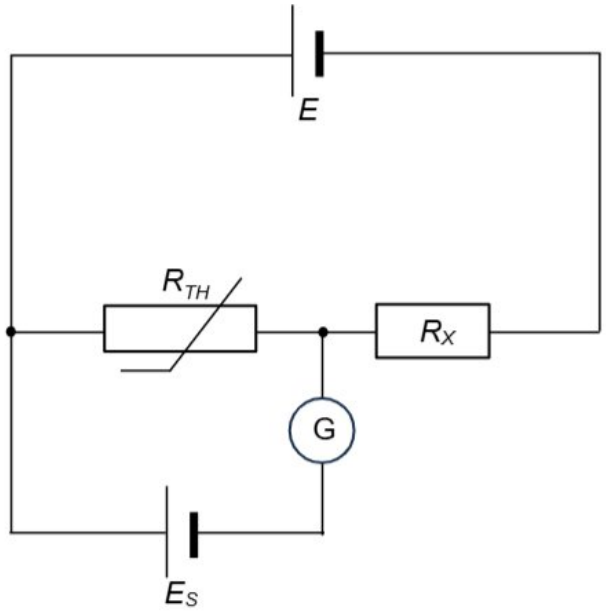


Fig. 5.2

At room temperature, there is no current detected in the galvanometer.

(A) Express E_S in terms of E at room temperature.

E_S = ... [2]

(B) State and explain whether the galvanometer will show any deflection if X is replaced with another resistor that has a lower resistance.

.....

.....

.....

..... [2]

(b) A piece of resistance wire BC of length 80.0 cm and diameter 0.027 cm has a resistivity of 4.9×10^{-7} Ω m.

(i) Show that the resistance of BC is 6.8 Ω.

[1]

(ii) Wire BC is now connected to a circuit, as shown in Fig. 5.3.

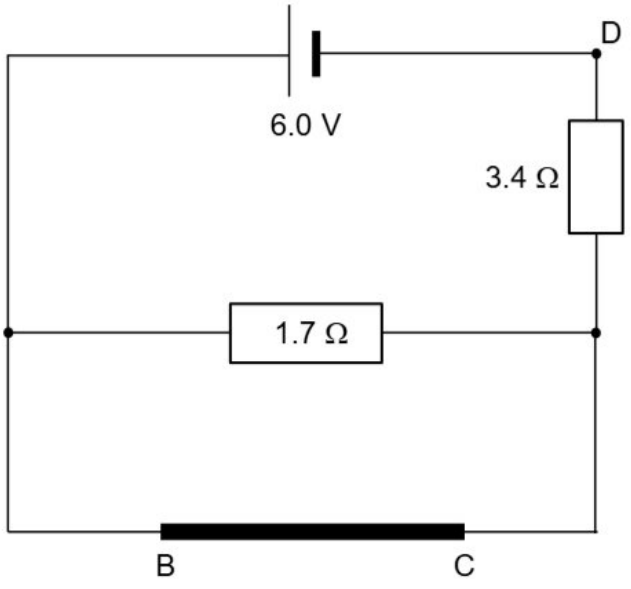


Fig. 5.3

(A) Calculate the current through the battery.

current = ... A [2]

(B) A voltmeter is connected to points B and D.

Determine the reading of the voltmeter.

voltmeter reading = ... V [1]

[Total:12]